BMSO603

DATA MINING AND PREDICTIVE ANALYTICS

Group 2 Final Project

Project Title: Predicting Obesity Risk Factors

ORIGINAL WORK STATEMENT:

We the undersigned certify that the actual composition of this proposal was done by us and is original work.

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Executive Summary

This study investigates how lifestyle behaviors and physical attributes influence obesity levels, using a dataset of 2,111 individuals that captures demographic, nutritional, and daily-activity patterns. The main goals of the study included identifying physical and behavioral features that were associated with a higher risk for obesity and determining which attributes play a bigger role in predicting individuals at risk.

To achieve this, we developed several predictive algorithms and compared them. These models were trained and tested after appropriate data preprocessing steps were taken.

Across the models, XGBoost produced the strongest predictive performance, hitting 97.9% for the combined model, showing that obesity risk can be reliably identified. Variables related to eating habits, such as calorie consumption, snacking habits, or vegetable intake, were consistently among the strongest predictors. These findings highlight that behavioral indicators rather than clinical characteristics may offer highly actionable insight for early obesity intervention strategies. Although biology and genetics are strong factors of an individual's risk for obesity, with preventative measures regarding their lifestyle choices, they can reduce their risk and outcome.

Overall, this study demonstrates that machine-learning models can be effectively used to classify obesity levels and identify high-risk profiles, supporting better-targeted wellness programs and preventive health strategies.

Data Description

The dataset used in this study is from Kaggle.

Full link: https://www.kaggle.com/datasets/ankurbajaj9/obesity-levels/data

The dataset contains behavioral, demographic, lifestyle, and dietary-pattern information for individuals from Mexico, Peru, and Colombia. The objective of the dataset is to estimate the obesity levels of individuals based on their lifestyle choices, physical activity habits, and eating behavior.

The numerical variables in the dataset include:

- Age
- Height
- Weight
- FCVC (Frequency of consumption of vegetables)
- NCP (Number of main meals)
- CH2O (Consumption of water daily)
- FAF (Physical activity frequency)

- TUE (Time using technology devices)

The categorical variables and their measurement in the dataset include:

- Gender: Male, Female
- Family_history_with_overweight: yes, no
- FAVC (Frequent consumption of high caloric food): yes, no
- CAEC (Consumption of food between meals): no, sometimes, frequently, always
- SMOKE: yes, no
- SCC (Calories consumption monitoring): yes, no
- CALC (Consumption of alcohol): no, sometimes, frequently, always
- MTRANS (Transportation used): public transportation, walking, automobile, motorbike, bike
- NObeyesdad (Weight Classification): Normal Weight, Overweight Level I, Overweight Level II, Obesity Type I, Obesity Type II, Obesity Type III, Insufficient Weight

Approximately 23% of the dataset represents real survey responses, while the remaining 77% was synthetically generated using controlled sampling and prediction techniques to preserve realistic distributions.

- Sample Size (n): 2,111 observations

- Number of Variables (k): 17 total variables (16 predictors + 1 target outcome)

Research Question

The questions we plan to investigate are:

1) Which physical attributes are at a higher risk of obesity?
    - This includes exploring distributions and patterns in age, height, weight, gender, and family history of overweight.
2) Can we predict obesity level by a person's lifestyle and behavior choices?
    - Models were built using variables such as physical activity frequency, fast-food consumption, emotional eating tendencies, water intake, etc. to see which impacted an individual's risk of obesity.
3) Are behavioral patterns stronger predictors of obesity than physical traits?
    - Combined features selected in the first two research questions to see which attributes, physical or behavioral, played a bigger role in determining one's risk of obesity.

Methodology

The project applied different machine learning and data mining techniques including data preprocessing, EDA (exploratory data analysis), feature engineering, and various classification models. Each model was evaluated based on its accuracy, RMSE (root mean squared error) and F1 score. The following techniques were essential to the study:

1. **Data Preprocessing and EDA**

   The dataset was prepared before running the models for analysis. The dataset did not contain missing values, therefore all observations remained in the analysis. Categorical features such as gender, FAVC and CH2O were encoded using one-hot encoding, so the models could process these variables numerically without imposing order. The dataset was split into training and testing sets to allow for generalization in the models and prevent overfitting. The main characteristics of the dataset were visualized including variable distributions and relationships between features. This allowed us to understand the variables and their descriptive statistics before feature selection for the models.

2. **Feature Engineering**

   Models were created for two separate groups to evaluate the impact of different features and their risk on obesity:

   - **Physical attributes:** Age, Height, Weight, Gender, Family History of Overweight

   - **Lifestyle and behavioral factors:** Physical activity, fast-food intake, emotional eating behavior, water consumption, etc.

   Obesity is influenced by both biological and behavioral factors. Separating the models allowed us to compare which type of variable carried stronger predictive power and to answer one of our core research questions more directly, “Are behavioral patterns stronger predictors of obesity than physical traits?”

3. **Model Selection**

   We selected four machine-learning models, each offering different strengths and interpretive benefits.

   Model 1: Multinomial Logistic Regression (Baseline)

   This model provides interpretable coefficients and odds ratios and serves as a baseline to compare against more complex models. It helps identify the direction and strength of the relationships between physical attributes and obesity classes. Although not expected to be the top performer, it offers valuable benchmarks and insight.

   Model 2: Decision Tree

The decision tree is highly interpretable and captures non-linear relationships and interaction effects between features. It also identifies important splitting variables (e.g., fast-food intake, activity level). We applied cost-complexity pruning to reduce overfitting/underfitting and improve generalization, which is especially useful with many behavioral categories. This model helped to determine feature importance through the tree splits by its root node and leaf nodes.

Model 3: Random Forest

Random Forest combines multiple decision trees to improve accuracy and reduce overfitting. It performs well with categorical and nonlinear behavioral data and provides stable feature importance rankings. Because it averages many trees, it typically outperforms a single tree and helped to evaluate how reliably we could predict obesity using different physical attributes.

Model 4: XGBoost + Hyperparameter Tuning

XGBoost is similar to Random Forest except it corrects the errors from the previous tree built sequentially. Combining it with RandomizedSearchCV allowed the model to tune its hyperparameters efficiently and find the model with the optimal parameters to maximize performance and accuracy.

4. **Evaluation Metrics**

We evaluated all models using:

- **Accuracy:** overall correctness
- **Macro F1 Score:** treats each obesity class equally, which is important due to class imbalance
- **RMSE:** used as an additional indicator of classification error

Multiple metrics were used because accuracy alone can be misleading in multi-class problems. The macro F1 score gives a clearer picture of whether the model is performing well across all obesity levels rather than just the most common classes.

This combination of preprocessing steps, modeling techniques, and evaluation metrics was chosen because the study is a multi-classification problem with many nonlinear relationships between features. Additionally, comparing multiple models provides deeper insight into the true predictors of obesity in individuals. Testing complex ensemble models can capture patterns and trends that other models might miss. Overall, this approach allowed us to build accurate models while still maintaining interpretability and addressing the complexity of obesity prediction.

Results and Findings

1. **Which physical attributes are at a higher risk of obesity?**

   The feature selection for physical attributes included an individual's gender, age, height, weight, and family history of obesity.

	Model 1: Multinomial Logistic Regression	**Model 2: Decision Tree**	**Model 3: Pruned Decision Tree (Cost-Complexity)**	**Model 4: Random Forest**
Accuracy	0.889	0.888	0.957	0.954
RMSE	0.709	0.825	0.337	0.594
F1 Score	0.887	0.881	0.955	0.951

Table 1: Evaluation of Models

Table 1 summarizes the results of the four models that were evaluated using accuracy, root mean squared error (RMSE) and F1 score. The pruned decision tree using cost-complexity pruning demonstrates the strongest performance with the random forest model coming in a close second. Model 3 has the strongest performance metrics with the lowest RMSE and high accuracy and F1 score, indicating its ability to correctly predict obesity levels while maintaining a strong balance between recall and precision. Model 4 had a higher RMSE compared to model 3 which means that on average, the predictions by model 4 are off by about 0.594 compared to 0.337.

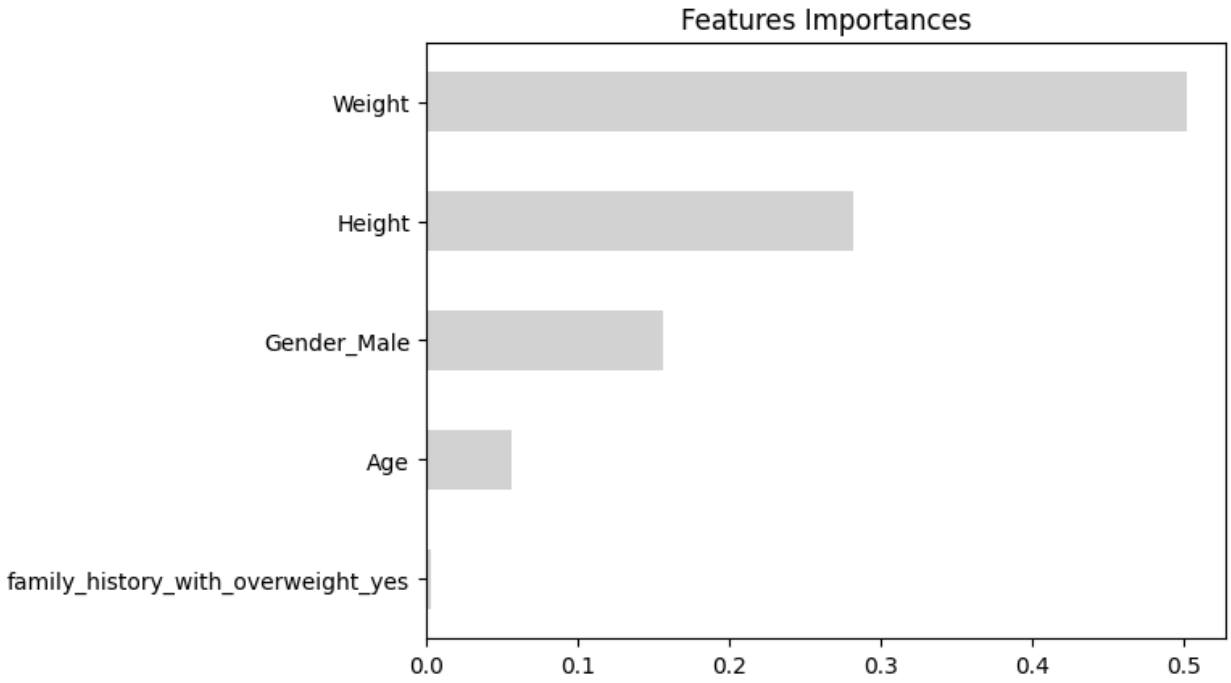

Figure 1: Features Importance of Pruned Decision Tree

According to the features importance chart in figure 1, weight has the highest importance which means it plays a significant role in predicting obesity levels. Specifically, from the decision tree in Figure 3 (Appendix), individuals who weigh over 74.58 kgs begin to have a higher risk for obesity with a classification of “obesity type I”. Shorter individuals under 1.59 m are at a higher risk although their weight may appear less than the average. This is due to their Body Mass Index (BMI). Lastly, having a family history of being overweight does not play a major role in predicting which individuals are at risk.

2. **Can we predict obesity level by a person’s lifestyle and behavior choices?**

A person’s lifestyle choices can have an impact on their health and risk for obesity.

	Model 1: Decision Tree	**Model 2: XGBoost**	**Model 3: XGBoost with RandomizedSearchCV**
Accuracy	0.674	0.733	0.7461
RMSE	1.990	1.702	1.72
F1 Score	0.669	0.729	0.740

Table 2: Evaluation of Models

From the evaluation of the models, XGBoost had the best performance metrics for predicting different obesity classes. The XGBoost model hyperparameters were tuned using RandomizedSearchCV, which provided the optimal parameters for efficient model accuracy. The model had a 74.6% accuracy and an RMSE score of 1.72. From the features importance chart in figure 2, the feature `CAEC_no` (consumption of food between meals) played an important role in predicting weight class for an individual. Other important features in the model included `FCVC` (frequency of consumption of vegetables), `TUE` (time using technology devices) and `NCP` (number of main meals consumed). This shows that eating habits of an individual help to differentiate between the different weight classes. `TUE` could be an important behavioral feature because it can correlate with an individual's sedentary activity. Typically, the more screen time an individual has means the less time they spend moving their bodies and being physically active throughout the day. Features that were not as important were `MTRANS` (transportation used). The dataset only captures one mode of transportation for individuals when there may be a variety in someone's day. Additionally, the dataset is not including the numerical time spent using a specific mode of transportation. For example, if the dataset captured how long individuals spent walking as their transportation mode, it may have more importance as a feature in the model. Overall, the key features for predicting obesity were an individual's eating habits.

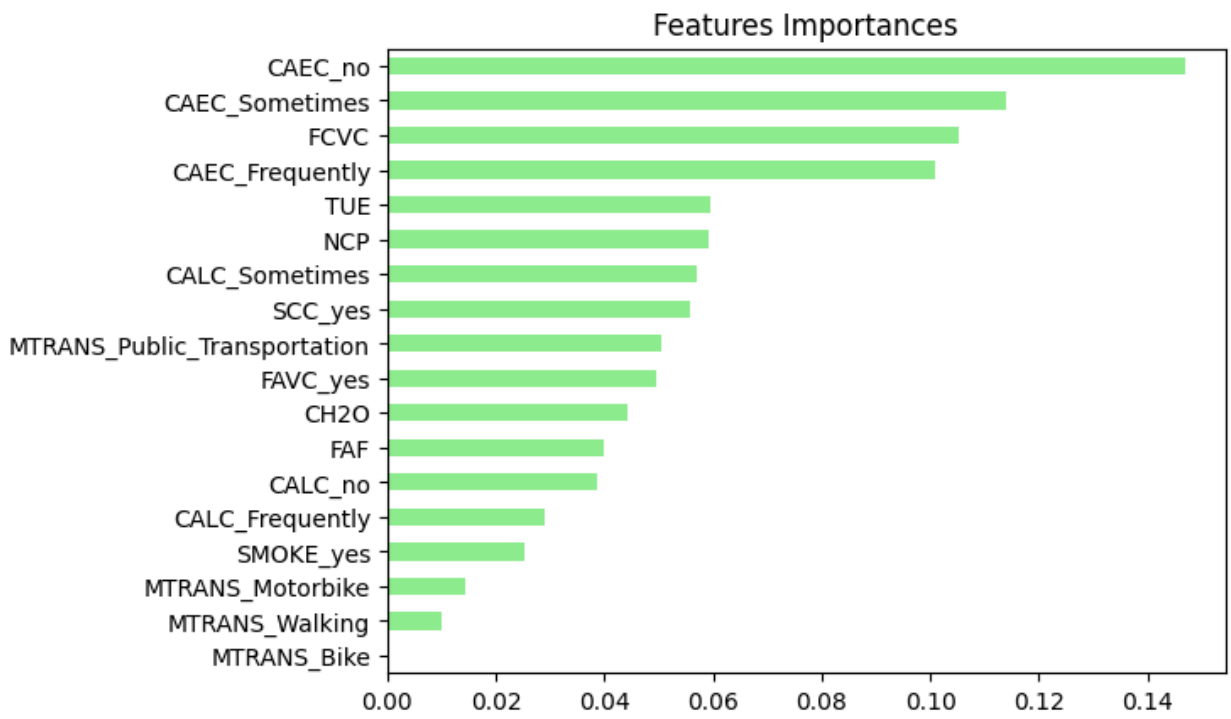

Figure 2: Features Importance of XGBoost

3. Are behavioral patterns stronger predictors of obesity than physical traits?

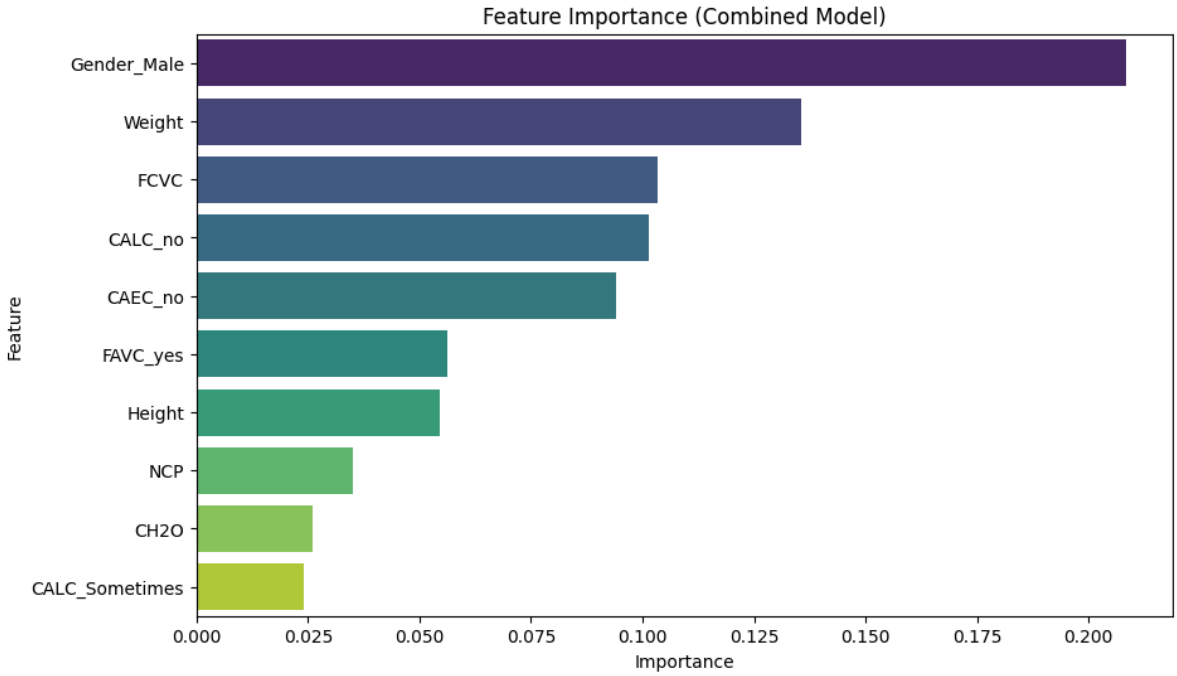

Figure 3: Feature Importance of Combined Model

To determine the most effective method for prediction, both physical attributes and lifestyle behaviors were tested in a comprehensive model. Three distinct algorithms were tested and compared to see which best captured the complex interactions between biology and behavior. Figure 3 illustrates the "Feature Importance" scores extracted from our final XGBoost model. This visualization provides a quantitative ranking of which variables the algorithm found most useful when distinguishing between the weight classes. `Weight` and `Gender` appear at the top of the hierarchy, meaning these features serve as the biological "anchors" for the model, adjusting for differences in body composition between men and women. Crucially, the model relies heavily on lifestyle variables to refine its predictions. Features such as `FCVC` (frequency of consumption of vegetables) and `CALC_no` (consumption of food between meals) carry significant predictive weight. The high ranking of these behavioral factors proves that biology cannot solely determine an individual's weight class. While weight and gender can serve as the baseline, an individual's risk category can be based on their habits (e.g., vegetable consumption versus technology usage). This validates our hypothesis that lifestyle data is essential for accurate high-precision classification (97.9%).

Model	Accuracy	Key Takeaway
Logistic Regression	~86.91%	Baseline. Fails to capture complex non-linear interactions
Random Forest	~95.6%	Strong. Effectively uses decision trees to map non-linear habits.
XGBoost (Winner)	~97.9%	Superior Performance. The gradient boosting approach effectively modeled the subtle edge cases between weight categories (e.g., "Overweight I" vs "Overweight II").

The victory of XGBoost (97.9%) over the linear baseline (~86.91%) provides a crucial business insight: Obesity is a non-linear problem. A linear model assumes risk increases at a constant rate with a positive or negative relationship between the variables. However, XGBoost captured an interaction effect between the features. For example, an individual that has high calorie intake (`FAVC`) might not lead to obesity if the individual also has high physical activity (`FAF`). This ability to learn conditional relationships and probabilities between physical and behavioral features makes XGBoost a practical candidate for real-world deployment.

Conclusion

This study demonstrates that obesity levels can be predicted with a strong accuracy using various data mining techniques. While physical attributes such as weight, height, and gender provide a solid baseline for predicting obesity rates, behavioral characteristics such as eating habits consistently emerged as the most reliable indicators of obesity risk. Across all models evaluated, XGBoost delivered the highest overall performance, followed closely by Random Forest. These models were especially effective at capturing the complex, non-linear relationships within behavioral data that simpler linear models like logistic regression could not fully detect.

The results also reveal important insights into the primary influences of obesity classification. Those with lower physical activity, frequent fast-food consumption, and higher emotional-eating tendencies were repeatedly classified into higher obesity categories. Also, variables such as caloric intake (`CALC`), vegetable consumption (`FCVC`), and meal frequency (`NCP`) played significant roles in the strongest-performing models. These findings reinforce that lifestyle choices serve as important predictive features when incorporated into machine-learning models.

Overall, the analysis confirms that obesity is strongly rooted in lifestyle patterns, and that data-driven modeling can support more targeted approaches to weight-related health outcomes. Lifestyle and behavioral variables offer high-impact, valuable targets for public-health interventions, preventative wellness programs, and personalized health assessments. By integrating behavioral and physical indicators, machine learning provides a strong framework for understanding obesity risk and learning effective health strategies outside of BMI.

Appendix (Optional)
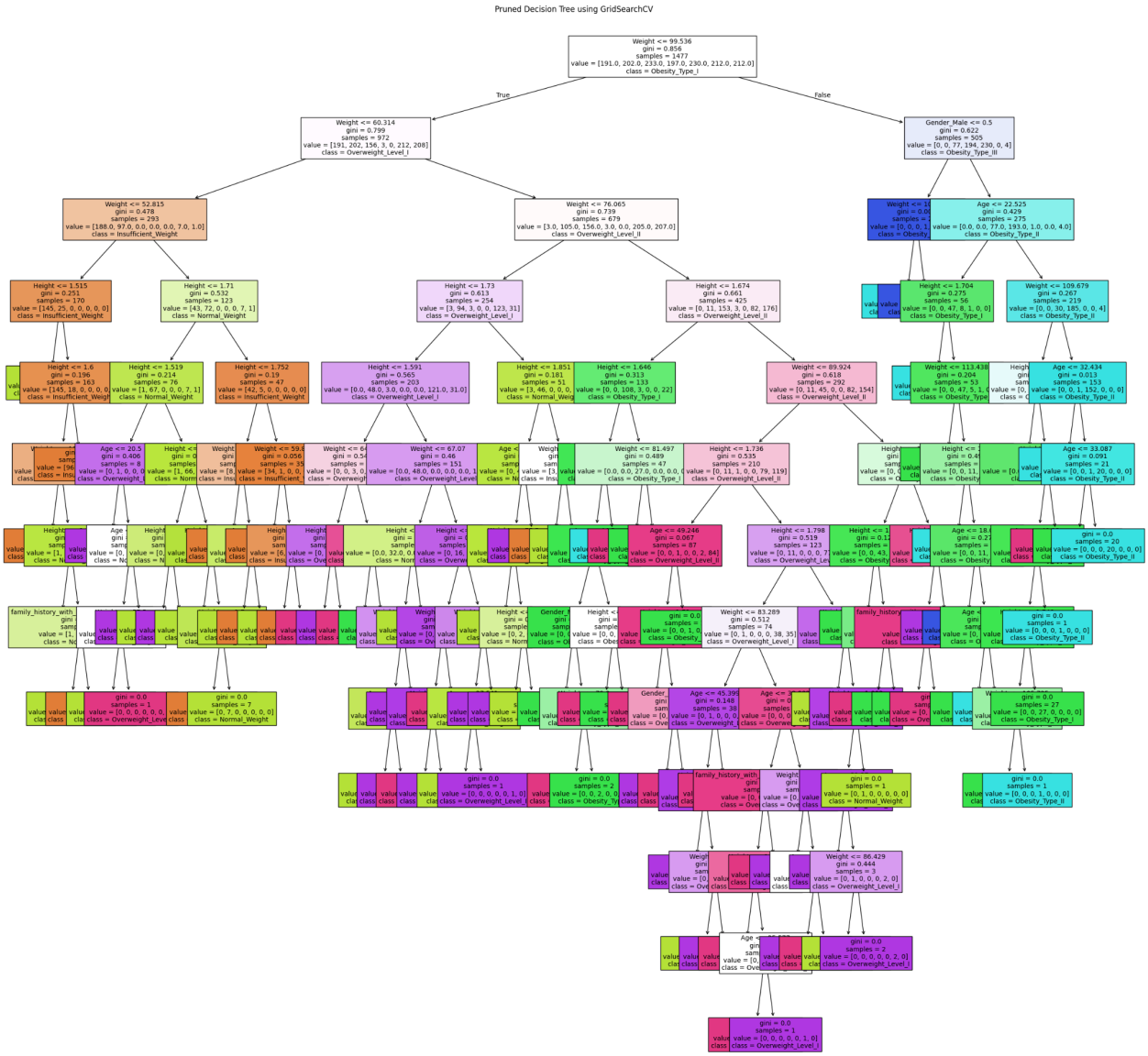
Figure 3: Pruned Decision (Cost-Complexity) for Research Question 1